

PHY 104 LECTURE NOTE

PRINCIPLE OF SUPERPOSITION

States that if two or more travelling waves are moving through a medium simultaneously, the resultant wave function at any point is the algebraic sum of the wave functions of the individual waves.

$$y_R(x, t) = y_1(x, t) + y_2(x, t) \quad (19)$$

$$y_R(x, t) = A \sin(kx - \omega t) + A \sin(kx - \omega t + \phi)$$

Using Trigonometric function, $\sin \alpha + \sin \beta = 2 \sin \frac{1}{2}(\alpha + \beta) \cos \frac{1}{2}(\alpha - \beta)$

$$y_R(x, t) = 2A \sin \frac{1}{2}(kx - \omega t + kx - \omega t + \phi) \cos \frac{1}{2}(kx - \omega t - kx + \omega t - \phi)$$

$$= 2A \sin \frac{1}{2}(2kx - 2\omega t + \phi) \cos \frac{1}{2}(-\phi) = [2A \cos \frac{1}{2}\phi] \sin(kx - \omega t + \frac{1}{2}\phi)$$

$$y_R(x, t) = [2A \cos \frac{1}{2}\phi] \sin(kx - \omega t + \frac{1}{2}\phi) \quad (20)$$

displacement magnitude oscillating term

Eqn. 20 is the resultant wave due to the interference of two (2) sinusoidal transverse waves, it is also a sinusoidal transverse wave with an amplitude and an oscillating term.

Note: Any waves that obey this principle is called **linear waves** and any waves that disobeys it, is called **non-linear waves**.

INTERFERENCE OF WAVES:

Is the combination of separate waves in the same region of space to produce a resultant wave.

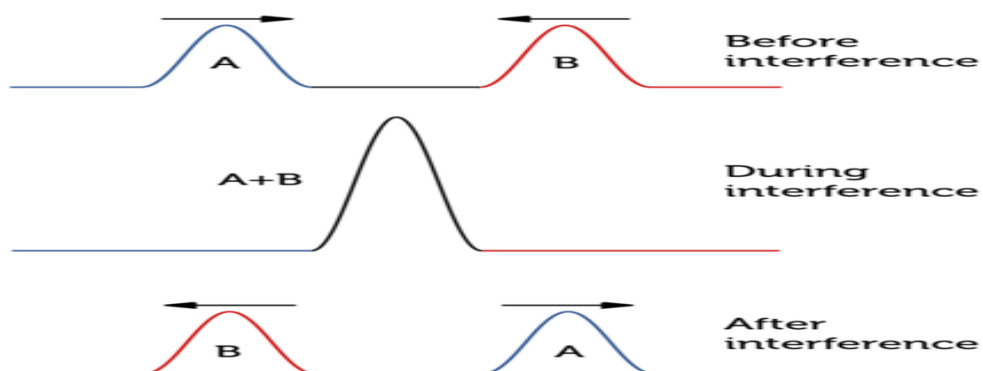


Fig. 5. Constructive interference of waves.

For the two (2) pulses shown in fig. 5, the displacement of the medium is in the positive y-direction for both pulses, and the resultant wave (created when the pulses overlap) exhibits a displacement greater than that of either individual pulse. Because the displacements caused by the two pulses are in the same direction, we refer to their superposition as **Constructive interference**.

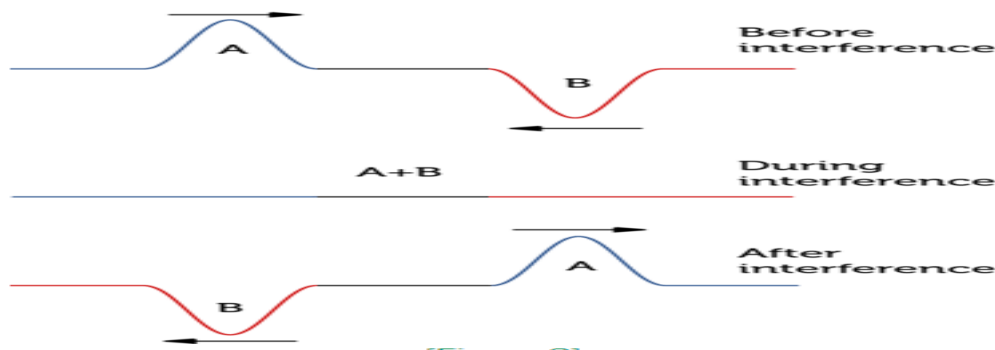


Fig. 6. Destructive interference of waves.

For the two (2) pulses shown in fig. 6 above, the displacement of the medium is in opposite y-direction where one pulse is inverted relative to the other. When the pulses begin to overlap, the resultant wave is given by $y_1 + y_2$ exhibits lesser displacement, and then, the two pulses pass through each other; however the displacements caused by the two pulses are in opposite directions, we refer to their superposition as **Destructive interference**.

1. Two waves are described by: $y_1 = 0.30 \sin[\pi(5x - 200t)]$

$$y_2 = 0.30 \sin \left[\pi(5x - 200t) + \frac{\pi}{2} \right]$$

Where y_1, y_2 and x are in meters and t is in seconds. When these two waves are combined, a traveling wave is produced. What are the (a) Amplitude, (b) Wave speed, and Wavelength of that traveling wave?

Soln.

$$y_R(x, t) = y_1(x, t) + y_2(x, t)$$

$$y_R(x, t) = 0.30 \sin[\pi(5x - 200t)] + 0.30 \sin \left[\pi(5x - 200t) + \frac{\pi}{2} \right]$$

$$y_R(x, t) = [2A \cos \frac{1}{2}\phi] \sin(kx - \omega t + \frac{1}{2}\phi)$$

$$y_R(x, t) = \left[2(0.3) \cos \frac{1}{2} \left(\frac{\pi}{2} \right) \right] \sin \left(5\pi x - 200\pi t + \frac{1}{2} \left(\frac{\pi}{2} \right) \right)$$

$$y_R(x, t) = \left[0.6 \cos \frac{\pi}{4} \right] \sin \left(5\pi x - 200\pi t + \frac{\pi}{4} \right)$$

$$y_R(x, t) = [0.6 \cos 45] \sin(5\pi x - 200\pi t + 45)$$

$$y_R(x, t) = 0.4243 \sin(5\pi x - 200\pi t + 45)$$

a) Amplitude $A = 0.4243\text{m}$

b) Wave speed $V = \frac{\omega}{k} = \frac{200\pi}{5\pi} = \frac{40\text{m}}{\text{s}}$

c) Wavelength, $\lambda = ?$

But Angular wave number, $k = \frac{2\pi}{\lambda}$.

$$\lambda = \frac{2\pi}{k}, \text{ where } k = 5\pi \text{ rad m}^{-1}$$

$$\lambda = \frac{2\pi \text{ rad}}{5\pi \text{ rad m}^{-1}} = 0.4\text{m}$$

DIFFRACTION:

Means the bending of waves around corners of an obstacle or aperture. Generally, diffraction depends on the aperture and wavelength of the wave.

The smaller the width of the aperture in relation to the wavelength of the wave, the more the diffraction of the wave. The larger the aperture's width compared to the wavelength of the wave, the less the diffraction of the wave or no diffraction will occur. For example, (1) diffraction of light can be seen by looking at a street light through a cloth umbrella, (2) sound waves are diffracted by doorways, hence we hear around the corner.

Diffraction refers to various phenomena that occur when a wave encounters an obstacle or slit.

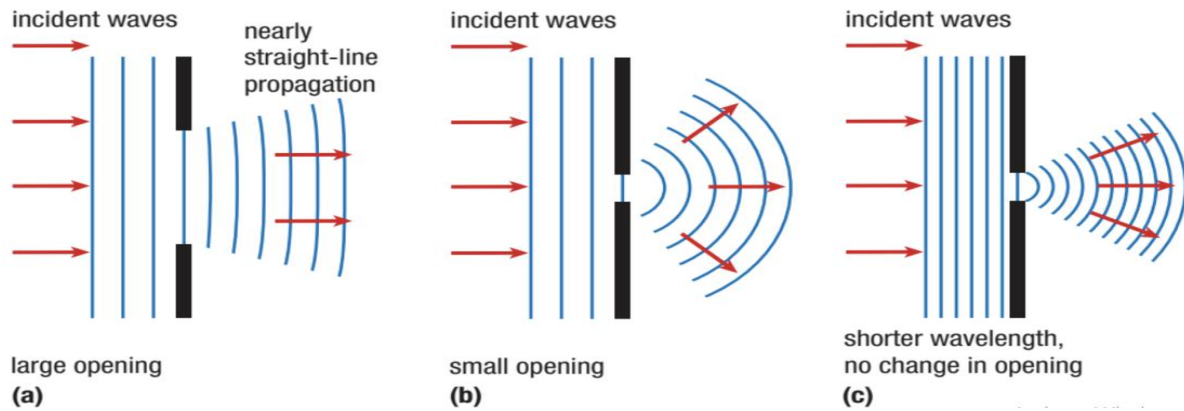


Fig. 7. (a) & (b) as the size of the slit (aperture) decreases, diffraction increases. (c) with a shorter wavelength and no change in the size of the opening, there is less diffraction.

POLARIZATION:

Polarization of waves is the confinement of the vibration of the particles of a wave to certain planes (in one direction) with the aid of a polarizer. There exists in the polarizer a certain characteristic polarizing direction which transmits on those wave-train components whose vibration are in agreement with the direction and absorb those not in agreement with the direction. The resulting wave is said to be polarized.

Polarization occurs only in transverse waves and not in longitudinal waves. Hence, polarization is a proof that light is a transverse waves.

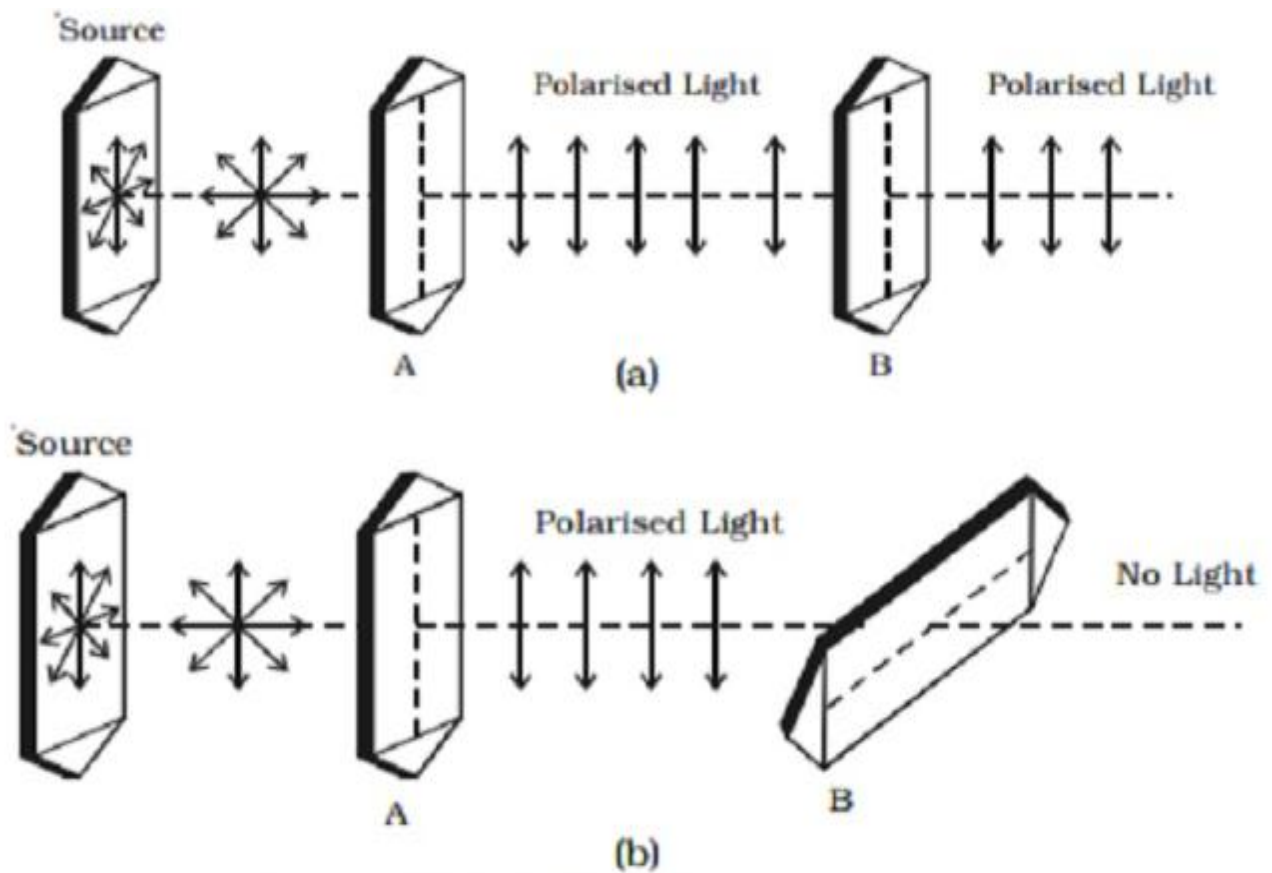


Fig. 8. Plane polarization of transverse wave (light).

VELOCITY OF LIGHT:

Velocity of light depends upon the medium through which it travels. The velocity of anything with mass is always less than the velocity of light in a vacuum. The velocity of light in a vacuum is a universal constant in all the frame of reference with value of $3.0 \times 10^8 \text{ m/s}$ approximately. It is denoted by C . $C = f\lambda$

Energy in Waves

The amount of energy in a wave is related to its amplitude and its frequency. Large-amplitude earthquakes produce large ground displacements. Loud sounds have high-pressure amplitudes and come from larger-amplitude source vibrations than soft sounds. Large ocean breakers churn up the shore more than small ones.

The energy of the wave depends on both the amplitude and the frequency. If the energy of each wavelength is considered to be a discrete packet of energy, a high-frequency wave will deliver more of these packets per unit time than a low-frequency wave. We will see that the average rate of energy transfer in mechanical waves is proportional to both the square of the amplitude and the square of the frequency. If two mechanical waves have equal amplitudes, but one wave has a frequency equal to twice the frequency of the other, the higher-frequency wave will have a rate of energy transfer a factor of four times as great as the rate of energy transfer of the lower-frequency wave. It should be noted that although the rate of energy transport is proportional to both the square of the amplitude and square of the frequency in mechanical waves, the rate of energy transfer in electromagnetic waves is proportional to the square of the amplitude, but independent of the frequency.

1. A stationary motion detector sends sound waves of frequency 0.150 MHz towards a truck approaching at a speed of 45.0 m/s . what is the frequency of the waves reflected back to the detector?
2. A stationary motion detector sends sound waves of frequency 0.150 MHz towards a truck approaching at a speed of 45.0 m/s . what is the frequency of the waves reflected back to the detector?

The total mechanical energy of the wave is the sum of its kinetic energy and potential energy. The kinetic energy $K = \frac{1}{2} M v^2$ (1) of each mass element of the string of length Δx is $\Delta K = \frac{1}{2} (\Delta m) V^2 y$ (2)

$\Delta K = 12(\Delta K) V^2 y$ (3) as the mass element oscillates perpendicular to the direction of the motion of the wave. Using the constant linear mass density, the kinetic energy of each mass element of the string with length Δx is

$$\Delta K = 12(\mu \Delta x) M v^2 y. \quad (4)$$

A differential equation can be formed by letting the length of the mass element of the string approach zero,

$$Dk = (\mu \Delta x) v^2 y = 12(\mu dx) v^2 y \quad (5)$$

$$\lim_{\Delta x \rightarrow 0}$$

Since the wave is a sinusoidal wave with an angular frequency ω , the position of each mass element may be modeled as

$$y(x,t) = A \sin(kx - \omega t) \quad (6)$$

Each mass element of the string oscillates with a velocity

$$V_y = \partial y(x,t) / \partial t = -A\omega \cos(kx - \omega t) \quad (7)$$

The kinetic energy of each mass element of the string becomes

$$dK = \frac{1}{2} (\mu dx) (A\omega \cos(kx - \omega t))^2 \quad (8)$$

$$= \frac{1}{2} (\mu dx) A^2 \omega^2 \cos^2(kx - \omega t) \quad (9)$$

The wave can be very long, consisting of many wavelengths. To standardize the energy, consider the kinetic energy associated with a wavelength of the wave. This kinetic energy can be integrated over the wavelength to find the energy associated with each wavelength of the wave.

SOUND WAVES

Sound waves are longitudinal waves and require a medium for their propagation or transmission (Mechanical waves). Solid, liquid and gases all transmit sound waves. Sound waves travel faster in a denser medium i.e. the velocity of sound waves varies from one medium to another and also varies with the temperature of the transmitting medium, $V = (331 \text{ m/s}) \sqrt{1 + \frac{T_c}{273^\circ\text{C}}}$ (e.g. speed of sound wave in wet or dry air is 330 m/s or 343 m/s respectively, speed of sound wave in liquid is about 1,480 m/s and speed of sound wave in solid medium is approximately 6,000 m/s). Therefore, sound waves do not travel in a vacuum (free space).

INTENSITY, PITCH AND QUALITY OF SOUND

- 1. SOUND INTENSITY:** is defined as the sound power per unit area flowing perpendicular to the direction of the wave. The unit is Watts/m².

$$\text{Mathematically, Intensity} = \frac{\text{Power}}{\text{Area}} = \frac{\text{Energy}}{\text{Area} \times \text{Time}} \quad (1)$$

Many sound intensity measurements are made relative to a standard threshold of hearing intensity, $I_0 = 10^{-12} \text{ W/m}^2$. The most common approach to sound intensity measurement

is to use the decibel scale: $\beta(\text{dB}) = 10 \log_{10} \left[\frac{I}{I_0} \right]$ (2)

Also, The intensity I is related to the displacement amplitude, A of the sound wave by: $I = \frac{1}{2} \rho V \omega^2 A^2$ (3)

Note that, the intensity of a sound wave is dependent on certain factors namely:

- i. It is directly proportional to the square of the amplitude of the wave. i.e. $I \propto A^2$
 - ii. It is directly proportional to the square of the angular frequency of the wave. i.e. $I \propto \omega^2$.
 - iii. It is directly proportional to the density of the air. i.e. $I \propto \rho_{\text{air}}$.
 - iv. It is inversely proportional to the square of the distance from the source of the sound. i.e. $I \propto \frac{1}{d^2}$
2. **PITCH:** Pitch of a sound or musical sound depends on the frequency of transmission of the sound waves or depends on the frequency of vibrations of the sound waves. Note that, a very high frequency gives rise to a very high pitch of a sound and vice-versa. The range of sound or audio frequency lies between 15 – 20,000 *Hz* for Humans.
- The pitch of a note is its position on the musical scale. It is a characteristic of a note which enables one to differentiate a high note from a low note.
3. **QUALITY OF SOUND:** describes those characteristics of sound which allows the ear to distinguish sounds which have the same pitch and loudness. The quality of a note therefore, depends on the overtones present in the note.
4. **LOUDNESS:** is the magnitude of the sensation resulting from a sound reaching the ear. Loudness of the sound depends on the intensity of the sound wave reaching the observer or listener concerned. It depends on the square of the amplitude of the sound wave i.e. a soft sound has a small amplitude but a loud sound has a large amplitude.

PROPAGATION OF SOUND IN SOLIDS, LIQUIDS & GASES

Propagation of sound in different media are as follows:

1. Solids: Velocity of sound wave in solids is given as:

$$V = \sqrt{\frac{E}{\rho}} \quad (4)$$

Where E = Young's modulus of the solid materials

$$\rho = \text{Density of the solid} = \frac{m}{V}$$

2. Liquids: Velocity of sound wave in liquids is given as:

$$V = \sqrt{\frac{B}{\rho}} \quad (5)$$

Where B = Bulk modulus of the liquid

$$\rho = \text{Density of the liquid} = \frac{m}{V}$$

3. Gases: Velocity of sound wave in gases is given as:

$$V = \sqrt{\frac{3RT}{m}} \text{ or } \sqrt{\frac{\gamma p}{\rho}} \quad (6)$$

Where R = molar gas constant ($8.3145 \text{ Jmol}^{-1}\text{K}^{-1}$) and $\gamma = \frac{C_p}{C_v}$

ρ = Density of the gas = $\frac{m}{V}$, m = molecular mass of the gas sample, T = Temperature in kelvin, γ = ratio of 2 specific heat capacity of gas, P = Pressure of gas, C_p = S.H.C of gas @ constant pressure, C_v = S.H.C of gas @ constant volume

Worked examples

1. Find the speed of sound in an Aluminium bar whose modulus is $7.0 \times 10^{10} \text{ N/m}^2$ for Aluminium and its density is $2.70 \times 10^3 \text{ kg/m}^3$.

Soln.

$$V = \sqrt{\frac{E}{\rho}} = \sqrt{\frac{7.0 \times 10^{10}}{2.70 \times 10^3}} = 5091.75$$

$$V = 5092 \text{ m/s}$$

2. What is the bulk modulus of a distilled water, if 32.0 g of distilled water occupies 22.4 litres and the speed of the sound in the distilled water is 1480 m/s.

Soln.

Mass, $m = 32.0 \text{ g} = 0.032 \text{ kg}$

Volume, $v = 22.4 \text{ litres} = \frac{22.4}{1000} \text{ m}^3 = 0.0224 \text{ m}^3$

Density, $\rho = \frac{m}{v} = \frac{0.032}{0.0224} = 1428.57 \text{ kg/m}^3$

$$V = \sqrt{\frac{B}{\rho}}$$

$$V^2 = \frac{B}{\rho}$$

$$B = V^2 \times \rho = (1480)^2 \times 1428.57 = 3.13 \times 10^9 \text{ N/m}^2$$

BEATS

Beats is the periodic variation in the loudness of a sound which is heard when two (2) notes of almost the same frequency are played simultaneously. If the frequencies of the sound waves are f_1 and f_2 , it can be shown that the beats frequency, f is given by; $f_{beat} = f_1 - f_2$. The beat frequency is the difference between the frequencies of the two (2) waves.

Beats are used in tuning of musical instrument e.g. a piano. Beats are also used to determine the frequency of a tuning fork.

Example 1.

Two tuning forks of frequency 206 Hz and 194 Hz are sounded simultaneously. Calculate the frequency of the resultant beat.

Soln.

$$f_{beat} = f_1 - f_2 = 206 - 194 = 12$$

$$f_{beat} = 12 \text{ Hz}$$

NOISE AND MUSIC: Noise is regarded as sound produced by the irregular vibration of an object so that there is no fixed frequency of vibrations. i.e. the frequency of vibration is not constant whereas, music or musical sounds are sound, which are originated from objects that vibrates uniformly with constant frequencies. i.e. they are produced by the regular vibration of an object so that there is fixed frequency of vibrations.

ECHO: Echo is caused by the reflection of sound waves when they meet a hard surface/objects. It is also the reflection of sound that arrives at the listener with a delay after the direct sound. This delay is proportional to the distance of the reflecting surface from the source and the listener.

Echoes are very useful in the following areas:

1. They are used to determine the depth of an ocean/sea through echo sounding.
2. They are used to detect cracks or faults in metal casting.
3. They are also used in oil/petroleum prospective.
4. They are used in military operations for the detection of enemies in sub-marines at sea level.

Disadvantages of Echoes are:

1. Echoes are not desirable in studios and auditorium.
2. Echoes cause disturbance while communicating through phones if it start echoing.

Methods/processes involved to minimize echo are namely:

1. Padding the walls of the studios with soft materials.
2. Making more windows in the studio/auditorium.
3. Ceilings and walls in a large auditorium are covered with perforated pads.

REVERBERATION: occurs as a result of perseverance of sound after the source of sound has cease/stop. e.g. thunder and multiple reflections in large room or hall.

ULTRASOUND, INTRASOUND & DOPPLER EFFECT

ULTRASOUND/ULTRASONIC: Sound waves of higher frequency than 20,000 *Hz* cannot be detected or heard by human beings but by certain animals e.g. dogs, cats, etc. can hear ultrasonic sound. Ultrasonic have short wavelength as compared with sound waves in audio frequency range.

Applications of Ultrasonic

- Industrial uses:
 1. They are used to kill bacteria in liquids during purification.
 2. They are also used in industry to locate or detect cracks or faults in metal casting.

For instance, Ultrasonic waves are sent into the metal being investigated and the beam reflected from the fault and crack is sent unto a cathode ray tube screen (oscilloscope) together with the reflection from the other end of the metal. The position of the crack and fault can then be easily detected.

- Medical uses:
 1. Ultrasonic are used to monitor the development of pregnancy in women. The ultrasonic waves are reflected back from the faults and in the way, the pregnancy can be monitored throughout the pre-natal period to check for any abnormalities.
 2. In diagnosis, Ultrasonic waves are reflected from inside a patient and then, amplified and displayed on a screen of a receiver. The image on the screen shows the various soft tissues and their interconnecting links and hence, the disease can be easily be detected.

INFRASOUND/INFRASONIC: Sound waves of very low frequencies than 20 Hz. Although, sounds of such low frequency are too low for humans to hear, but animals e.g. whales, elephants and other animals can detects infrasound and also use it to communicate. It can be used to detect volcanic eruptions.

WAVES IN PIPES, STRINGS AND RODS

When both ends of the pipe are open, it is called as open pipe but when one end is closed and one end is open, it is called a closed pipe.

When air vibrates inside a pipe, the waves produced are reflected at the ends producing a longitudinal stationary wave along the length of the pipe. Since, the air particles at the closed end of the pipe cannot move, there is always a node at this end. At the open end, the air particles are free to vibrate with maximum amplitude. An antinode is therefore, always formed at the open end.

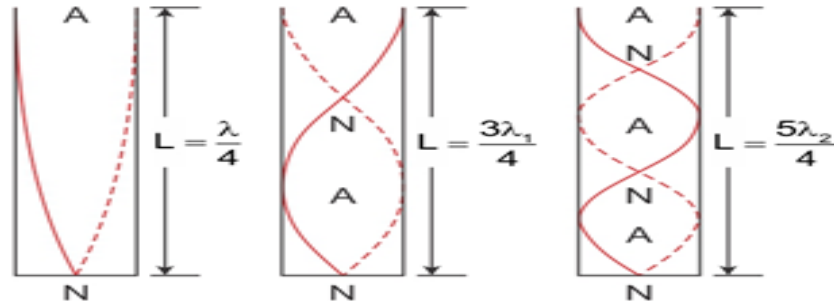
Vibrations produced in closed pipes

The simplest mode of vibration produced in a closed pipe is illustrated in fig. 11. The diagram shows the stationary wave formed in the air column when we blow gently at the open end. The disturbance created at this open end travels through the air column and is reflected at the closed end. There is therefore, a node (N) at the closed end and an antinode (A) at the open end. Since, the distance between a node and a consecutive antinode in a waveforms is $\frac{\lambda}{4}$,

We have from the diagram that the length of the tube: $l = \frac{\lambda}{4}$ or $\lambda = 4l$. Therefore, the fundamental frequency, f_0 is given by;

$$f_0 = \frac{V}{\lambda} = \frac{V}{4l}$$

$$f_0 = \frac{V}{4l} \quad (1)$$



By varying the length of air column in the tube i.e. increasing the length further by running off more water, a second position will be reached when a large sound will once again be heard. In this position, a second nod will be formed such that the wavelength of the vibrating air is now $l = \frac{3\lambda}{4}$ or $\lambda = \frac{4l}{3}$

The first overtone frequency, f_1 is given by;

$$f_1 = \frac{V}{4l/3} = \frac{3V}{4l} = 3f_0 \quad (2)$$

By further increasing the length, we can find the next overtone frequency with three nodes and so on. These are $5f_0, 7f_0 \dots$ Hence, the possible harmonics of a closed pipe are $f_0, 3f_0, 5f_0, 7f_0 \dots$ Note that only odd harmonics are present.

A harmonic is a note with frequency equal to an integral multiple of that of the fundamental note.

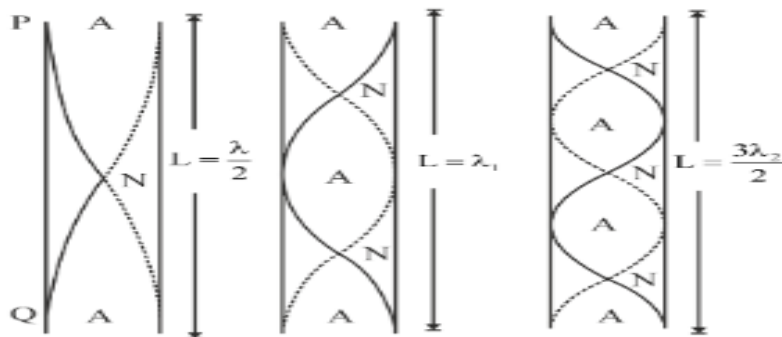
Vibrations produced in open pipes

In an open pipe, since the two ends of the pipe must end in an antinode, the fundamental or the simplest mode of the vibration (first harmonic) is that in which the mid-point is a node. Therefore,

$$l = \frac{\lambda}{2} \text{ or } \lambda = 2l$$

The fundamental frequency in this case is $f_0 = \frac{V}{\lambda} = \frac{V}{2l}$

$$f_0 = \frac{V}{2l} \quad (3)$$



The second harmonic has frequency, $f_1 = \frac{v}{\lambda} = \frac{v}{l} = \frac{2v}{2l} = 2f_0$ (4)

The third harmonic has frequency, $f_2 = \frac{v}{\lambda} = \frac{3v}{2l} = 3f_0$ (5)

Hence, the harmonics present in an open pipe are $f_0, 2f_0, 3f_0, 4f_0, \dots$ thus, all harmonics are possible.

If we are provided with only one tuning fork and a long pipe, it is possible to determine v by obtaining both the first and second positions of resonance with the same fork.

For the first position; $l_1 + c = \frac{\lambda}{4}$ (6)

Where c is the end correction which arises from the fact that the antinode at the top does not exactly coincide with the top of the tube.

For the second position; $l_2 + c = \frac{3\lambda}{4}$ (7)

Equating (6) and (7) together in order to eliminate c ,

$$l_2 - l_1 = \frac{3\lambda}{4} - \frac{\lambda}{4} = \frac{\lambda}{2}$$

$$l_2 - l_1 = \frac{\lambda}{2}$$

$$2(l_2 - l_1) = \lambda$$

$$\lambda = 2(l_2 - l_1) \quad (8)$$

The speed of sound, $v = f\lambda = 2f(l_2 - l_1)$ (9)

Worked examples

1. In a resonance tube experiment, a tuning fork of frequency 256 Hz gave resonance when the water level was 32.5 cm below the open end of the tube. If the next position of resonance was 100 cm , what is the speed of sound in air?

Soln.

$$1^{\text{st}} \text{ position; } l_1 = \frac{\lambda}{4}$$

$$2^{\text{nd}} \text{ position; } l_2 = \frac{3\lambda}{4}$$

$$l_2 - l_1 = \frac{\lambda}{2}; f = \frac{v}{2(l_2 - l_1)} \text{ and } v = 2f(l_2 - l_1)$$

$$f = 256 \text{ Hz, } l_1 = 0.325 \text{ m, } l_2 = 1.00 \text{ m}$$

$$v = 2(256)(1.00 - 0.325) = 345.6$$

$$v = 345.6 \text{ m/s}$$

2. Calculate the frequency of the first overtone of a closed pipe of length 20 cm, if the speed of sound in air is 340 m/s.

Soln.

$$l = 20 \text{ cm} = 0.20 \text{ m, } V = 340 \text{ m/s}$$

$$\text{Fundamental frequency for a closed pipe, } f_0 = \frac{V}{4l}$$

$$\text{First overtone for a closed pipe, } f_1 = \frac{3V}{4l} = 3f_0 = \frac{3 \times 340}{4 \times 0.20} = \frac{1020}{0.80} = 1275$$

$$f_1 = 1275 \text{ Hz}$$

3. The length of the vibrating air column of a simple resonance tube can be altered by adjusting a water level. Resonance is found for a fork of frequency 440 Hz, when the length of the air column is 18.8 cm and again when it is 57.3 cm. Explain how each of the positions occur; calculate the speed of sound in the air in the tube.
4. The frequency of the fundamental note of a closed pipe is 200 Hz. What is the frequency of the next possible note from the same pipe?

REFLECTION

Reflection of light is when light strikes a perfectly reflecting surface such as mirror for example, all the light energy is reflected. Therefore, reflection can simply be defined as the re-propagation of light waves when incident at a particular angle on a plane surface.

There are two (2) types of reflections namely are:

1. Smooth/regular reflection
2. Diffused/irregular reflection

Smooth/regular reflection: is produced only on smooth surfaces. The rays are reflected and the moment they strike smooth surfaces reflected rays are produced.

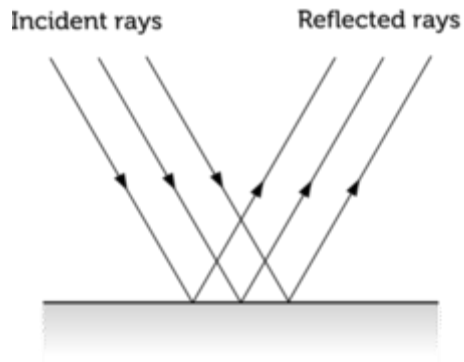


Fig. 1 Smooth/regular reflection.

Diffused/irregular reflection: is produced from rough surfaces. The rays striking the rough surfaces are scattered or reflected in different angles.

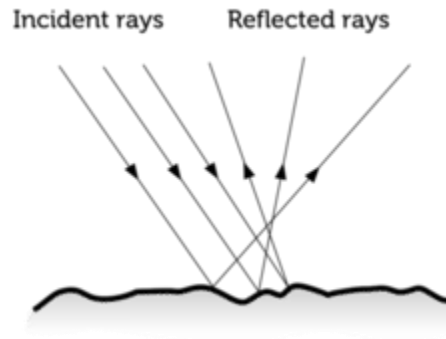


Fig. 2 Diffused/irregular reflection.

PLANE AND CURVED SURFACES

Plane surface is simply a flat (2 dimensional) surface. In theory, it has no edges and so it's infinite. Examples are plane mirrors, piece of paper, glass etc. Curved surface is also called a spherical surface i.e. it has edges and so it's finite. Examples are spherical mirrors, lenses etc.

LAWS OF REFLECTION

There are two (2) laws associated with reflection namely are:

- ✓ The 1st law of reflection states that the incident ray, the reflected ray, and the normal at the point of incidence, all lies on the same plane.
- ✓ The 2nd law of reflection states that the angle of incidence is equal to the angle of reflection. i.e. $\hat{i} = \hat{r}$

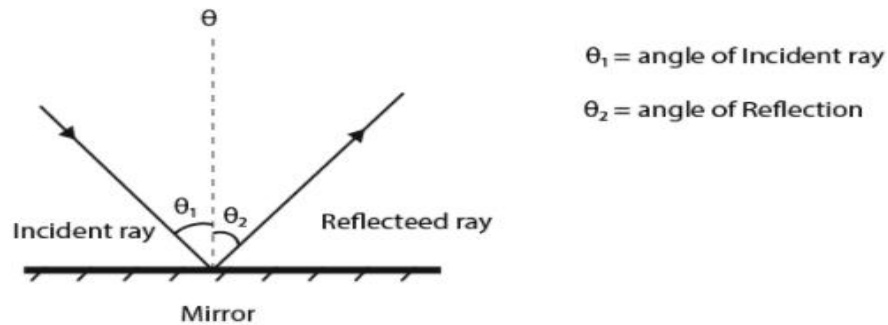


Fig. 3 Laws of reflection

Worked examples

1. A light ray strikes a reflective plane surface at an angle of 56° with the surface. Find:
 - (a) the angle of incidence
 - (b) the angle of reflection
 - (c) the angle made by the reflected ray and the surface
 - (d) the angle made by the incident and reflected rays.

Soln.

With the aid of a diagram below;

- (a) Angle of incidence, $\hat{i} = 90^\circ - 56^\circ = 34^\circ$
- (b) Angle of reflection, $\hat{r} = \hat{i} = 34^\circ$ (by the laws of reflection)
- (c) $q = 90^\circ - \hat{r} = 90^\circ - 34^\circ = 56^\circ$
- (d) $\hat{i} + \hat{r} = 34^\circ + 34^\circ = 68^\circ$

REFLECTION AT PLANE SURFACES

Every image produced from the plane surfaces are laterally inverted (opposite), the distance between the object and the plane. But the surfaces to the image remains the same. Thus, every image produced on a plane surface is virtual.

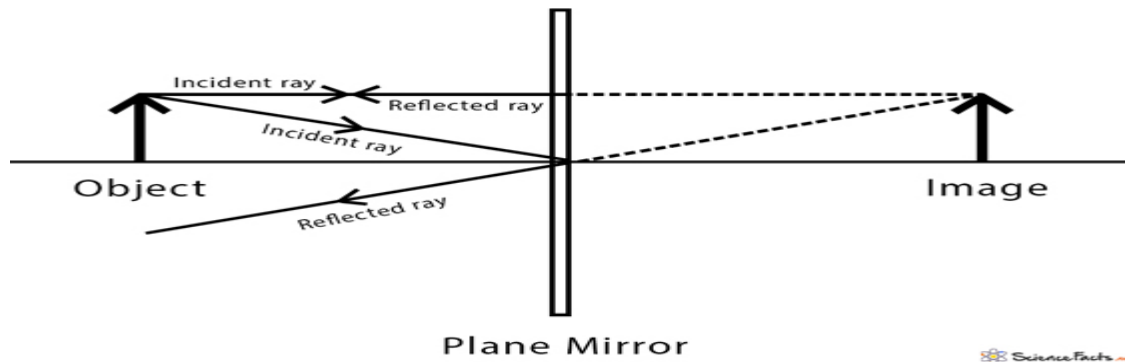


Fig. 4 Images produced on a plane mirror.

Summary:

- i. The image is the same as the object.
- ii. It is laterally inverted.
- iii. It is virtual i.e. cannot be formed on a photographic screen.

Properties of a Plane mirror

- i. Plane mirrors produce images with a number of distinguishable characteristics.
- ii. Images formed by plane mirrors are virtual, upright, left-right reversed.
- iii. Images formed by plane mirrors have the same distance from the mirror as the object distance.
- iv. Images formed by plane mirrors have the same size as the object.

Images produced on an inclined plane: Most inclined plane surfaces produced limited number of images, depending on their angle of inclinations. Two surfaces inclined at an angle of 90° produce three (3) different images while surfaces inclined at angle of 40° produces eight (8) different number of images.

Mathematically, Number of images, $n = \frac{360}{\theta} - 1$ (1)

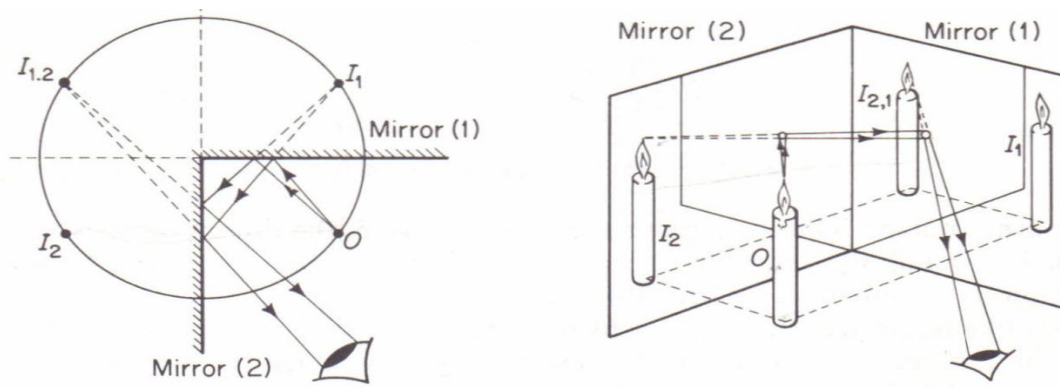


Fig. 5 Reflection from two (2) mirrors inclined at 90° .

Images produced from parallel plane surfaces: the number of images produced on a parallel surfaces is unlimited but sometimes, it could be obstructed, depending on the position of the observer. The images produced are laterally inverted.

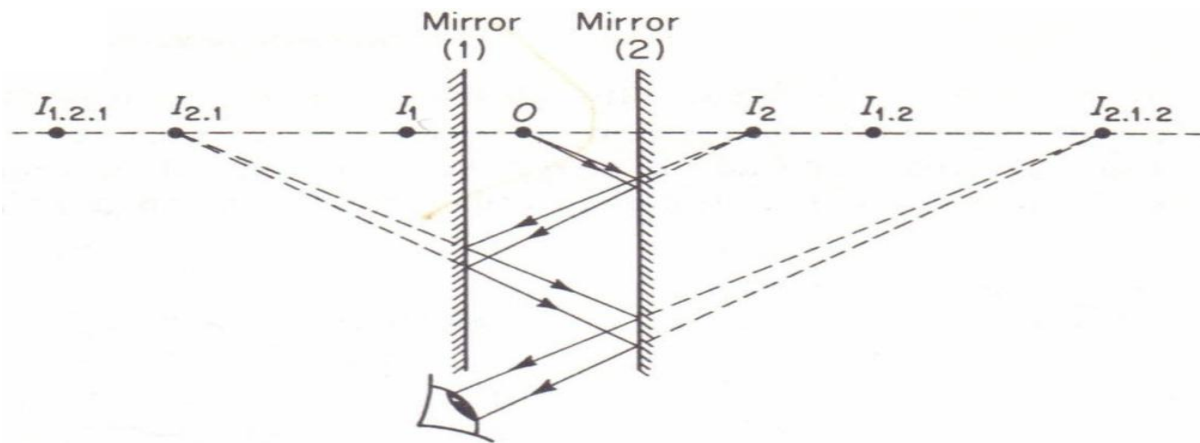


Fig. 6 Images formed by two parallel mirrors.

Applications of a Plane mirror

1. Plane mirrors are used in day-to-day activities as a viewing glass etc.
2. It is also used in making Periscope, Kaleidoscope, etc.
3. It can be used as Sextant device.

SPHERICAL MIRRORS

Spherical mirrors are also called curved surfaces and there are two (2) types of spherical mirrors which are determined by the size of the curved mirror and the images produced.

The two (2) types of spherical mirrors are concave and convex mirrors.

1. Concave mirror: this is a curved mirror that produces a real image and has the left-hand side (L.H.S) of its surface coated. The reflecting surface bends inwards, causing incident rays to converge after reflection.

Fig. 7 Concave mirror

2. Convex mirror: this is a curved mirror that produces a virtual image and has its right-hand side (R.H.S) of its surface coated. It causes incident rays to diverge after reflection. The reflecting surface bends outwards.

Fig. 8 Convex mirror.

Application of Spherical/curved mirror

1. Concave mirrors are used as shaving mirror and dentist mirror.
2. Convex mirrors are the most useful of all the curved mirrors available in areas of security, and Car driving mirrors.

MIRROR FORMULAE

The general mirror formula is, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ (2)

Where v = image distance

u = object distance

$f = \frac{r}{2}$ = focal length.

Linear magnification is defined as the ratio of the image height to the object height or the ratio of the image distance to the object distance.

$$M = \frac{H_z}{H_o} = \frac{v}{u} \quad (3)$$

Worked examples

1. An object is placed 20 cm in front of a concave mirror of focal length 12 cm, find the position and nature of the image.

Soln.

$u = -20 \text{ cm}$ (Object is placed to the left of the mirror)

$f = -12 \text{ cm}$ (Focal length of concave mirror)

$v = ?$

By using mirror formula, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$$\frac{1}{v} + \frac{1}{-20} = \frac{1}{-12}$$

$$\frac{1}{v} = \frac{1}{-12} + \frac{1}{20} = \frac{-8}{240} = -0.033$$

$v = -30 \text{ cm}$

The position of the image is -30 cm .

The negative sign indicates that the image is formed at the same side as that of object. The image distance is 30 cm . Thus, it is formed beyond the centre of curvature and the nature of the image is real, inverted and magnified.

2. An object is placed at a distance of 25 cm from a mirror. The mirror produces an image that is 20 cm , (i) in front the mirror and (ii) behind the mirror. For each case, find the focal length of the mirror and the nature of the mirror.

Soln.

The object is placed at the distance of 25 cm from the mirror i.e. $u = -25\text{ cm}$

The image is produced at 20 cm .

- i. In the first case; the image is in front of the mirror. Thus, the image is on the same side as that of an object. $v = -20\text{ cm}$.

Using the mirror formula; $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$$\frac{1}{f} = \frac{1}{-20} + \frac{1}{-25} = \frac{-45}{500} = -0.011$$

$$\frac{1}{f} = -0.011$$

$$f = -11.0\text{ cm}$$

Therefore, when the image is formed in front of the mirror, the focal length of the mirror is 11 cm . As the focal length is $-ve$, the mirror is concave mirror.

- ii. In second case: the image is formed behind the mirror at 20 cm from the mirror. Thus, the image is to the right side of the mirror. $v = 20\text{ cm}$.

Using the mirror formula; $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$$\frac{1}{f} = \frac{1}{20} + \frac{1}{-25} = \frac{-5}{-500}$$

$$f = 100\text{ cm}$$

Therefore, when the image is formed behind the mirror, the mirror is convex mirror because its focal length is always $+ve$. The mirror is a convex mirror.

REFRACTION

Refraction of light is defined as the property of a change in the direction of light, when it passes from one medium to another. In the process of refraction, the direction to which the ray of light follows changes as well as the velocity but the frequency remains the same along the line of propagation.

LAWS OF REFRACTION

There are two (2) laws associated with refraction namely are:

- ✓ The 1st law of refraction states that, the incident ray, the refracted ray and the normal at the point of incidence, all lies on the same plane.
- ✓ The 2nd law of refraction is also known as *Snell's law* which states that the ratio of sine of angle of incidence to the sine of the angle of refraction is a constant for a given pair of media. i.e. $\eta = \frac{\sin i}{\sin r}$ (1)

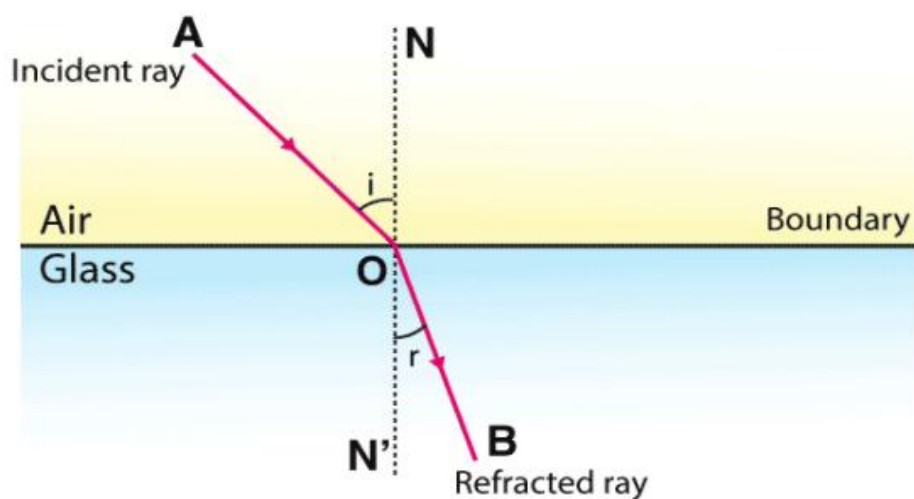


Fig. 9 The law of refraction

Refractive index η is a constant used to express the nature of medium through which light ray is propagated. The refractive index determines the extent of permeability at which ray travels through medium.

$$\eta = \frac{\sin \hat{i}}{\sin \hat{r}} = \frac{\text{Real depth}}{\text{Apparent depth}} \quad (2)$$

Refraction through Prisms

There are two (2) types of prisms namely are:

1. Rectangular prism
2. Triangular prism

Rays are produced from a rectangular prism which has the angle of refraction either produced from the normal or produced to the normal but rays transmitted through a triangular prism are always refracted towards the normal and close to the base of the triangular prism.

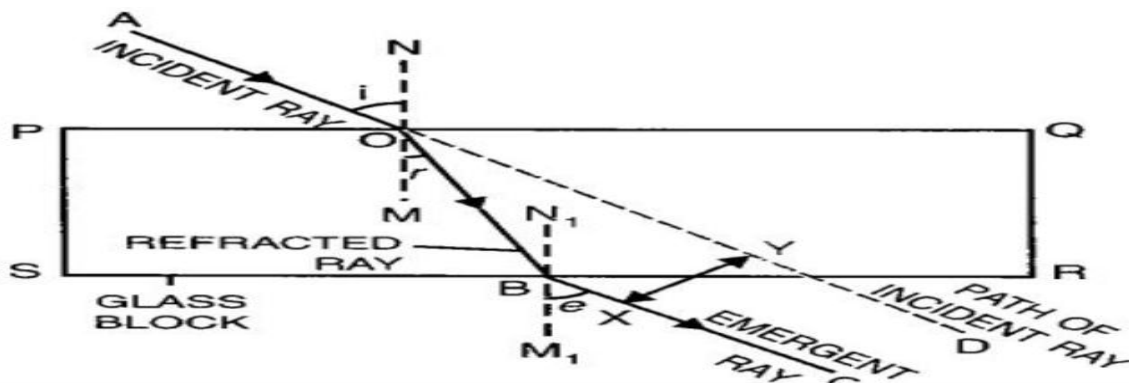


Fig. 10 Refraction by a Rectangular prism.

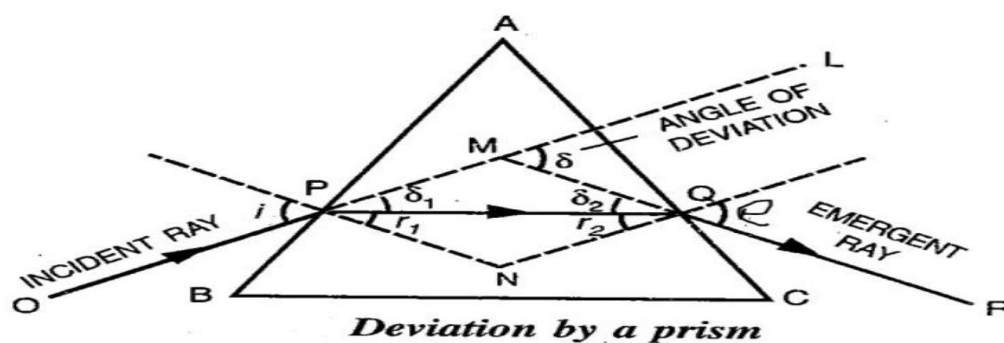


Fig. 11 Refraction by a Triangular prism.

Note:

Minimum deviation of the light d_m occurs when the ray passes symmetrically through the prism.

$${}_a\eta_g = \frac{\sin \hat{i}}{\sin \hat{r}}$$

$$i = \frac{A + d_m}{2}$$

$$r = \frac{A}{2}, A = 2r$$

Also, $A + d_m = 2i, d_m = 2i - A$

Substitute the values of i and r into ${}_a\eta_g$ equation.

$${}_a\eta_g = \frac{\sin\left(\frac{A+d_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} \quad (3)$$

REFRACTION THROUGH LENSES

Recall that, refraction simply means the change in speed of rays of light from one medium to another. Therefore, lenses are a transparent medium which could be glass, plastic and nylon which are bounded by non-coated spherical surface or curved surface.

There are two (2) known types of lenses namely are concave (diverging) lens and convex (converging) lens.

1. Concave lenses are biconcave, plano-concave, and diverging meniscus. The rays that passes through them are spread out i.e. diffused because they are thinner at the middle than the edge. They produce virtual, diminished and erect images.

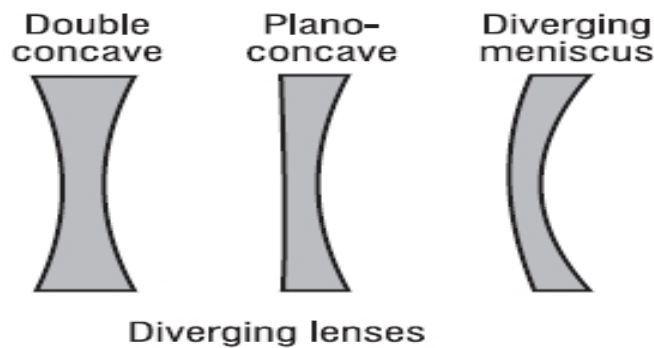


Fig. 12 Concave (diverging) lenses.

2. Convex lenses are also biconvex, plano-convex, and converging meniscus. The rays that passes through them comes together at the point of the focus, because they are thicker at the middle than the edge. They produce real, magnified and erect or inverted images.

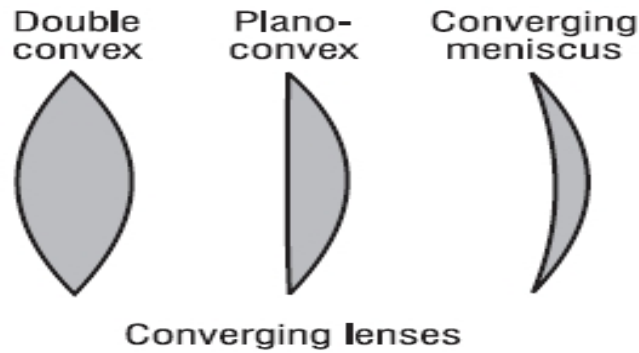


Fig. 13 Convex (converging) lenses.

Power of lenses

The power of a lens is the ratio of one to the focal length or the ratio of a focal length of a lens.

Mathematically, Power of a lens $P = \frac{1}{f}$ (4)

S.I unit is *Dioptre* (cm^{-1})

Note that, the power of a concave lens is negative while the power of a convex lens is positive.

TOTAL INTERNAL REFLECTION

Total internal reflection is the phenomenon in which there is a complete reflection of a wave incident at a sufficiently oblique angle on the interface between two media of which the second (external) medium is transparent to such waves.

Refraction is generally accompanied by partial reflection. When a wave train is refracted from a medium of higher refractive index to a medium of lower refractive index, the angle of refraction is greater than the angle of incidence. Hence, as the angle of incidence approaches a certain limit called the critical angle, the angle of refraction approaches 90° , at which the refracted ray becomes tangential to the interface. As the angle of incidence increases beyond the critical angle, the conditions of refraction can no longer be satisfied; so we have no refracted ray, and the partial reflection becomes total.

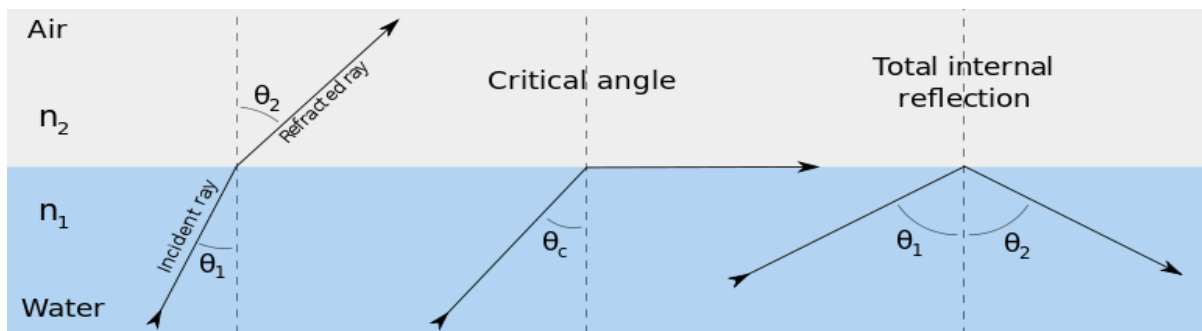


Fig. 14 Critical angle and total internal reflection

Applications of Total internal reflection:

1. Optical fibre works on the principle of total internal reflection.
2. The brilliance of a diamond is due to total internal reflection.
3. The phenomenon of mirage can be explained on the basis of total internal reflection.
4. The phenomenon of total internal reflection of light is used in many optical instruments like telescopes, microscopes, binoculars, spectroscopes, periscopes etc.

DISPERSION

Dispersion of light is the phenomenon of splitting of a beam of white light into its seven constituent colours when the white light passes through a transparent medium. It was first discovered by Sir Isaac Newton in the year 1866. Newton discovered that light is made up of seven (7) different colours. So he experimentally passed a beam of sunlight through a glass prism and the light is split by the prism into a band of 7 colours on his wall. He then, called this band of colours the 'spectrum'.

The order of colours from the lower end of the spectrum is violet (V), indigo (I), blue (B), green (G), yellow (Y), orange (O), and red (R). The sequence of these 7 colours so obtained in a spectrum can be remembered by using the acronym 'VIBGYOR'.

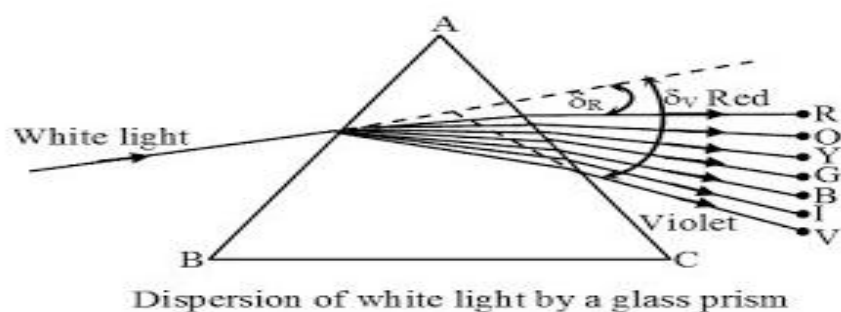


Fig. 15 showing the dispersion of light.

Cause of dispersion of light

The cause of dispersion of light is that white light consists of 7 different colours and each colour has different angle of deviation. Therefore, on passing through the prism different colours deviate through different angles. Hence, the 7 different colours of white light separates and form a spectrum.

Out of the 7 different colours, red colour deviates the least and hence the red colour is present at the top of the spectrum. On the other hand, violet colour deviates the most i.e. why violet colour is present at the lower end of the spectrum.

SPECTRA

Spectra (plural form of spectrum), the word was first used scientifically in Optics to describe the rainbow of colours in a visible light after passing through a prism. As scientific understanding of light advanced, it was then applied to the entire electromagnetic spectrum.

Electromagnetic waves exist with an enormous range of frequencies. This continuous range of frequencies is known as the electromagnetic spectrum. The entire range of the spectrum is often broken into specific regions. The sub-dividing of the entire spectrum into smaller spectra is done mostly on the basis of how each region of electromagnetic waves interacts with matter.

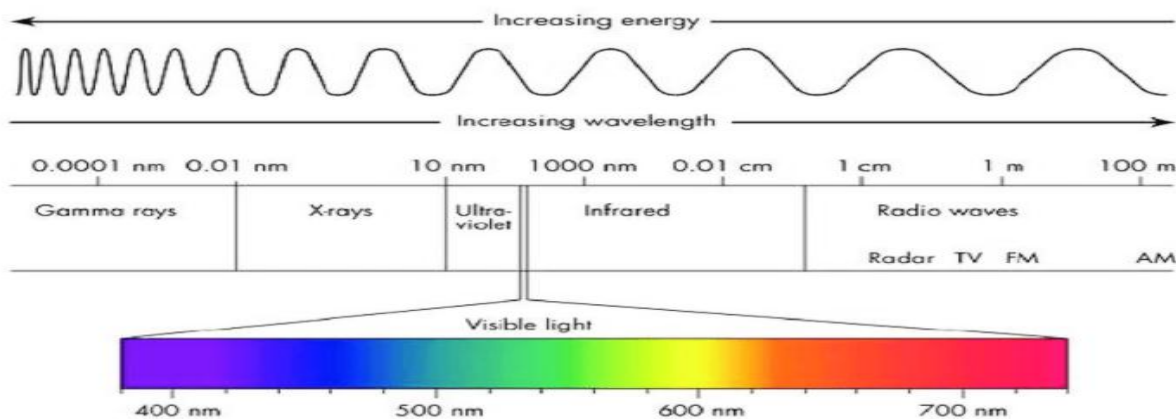


Fig. 16 showing the electromagnetic spectrum.

The diagram above depicts the electromagnetic spectrum and its various regions. The longer wavelength, lower frequency regions are located on the far left of the spectrum and the shorter wavelength, higher frequency regions are on the far right.

Spectrometer: is a scientific instrument used to separate and measure spectral components of a physical phenomenon. Spectrometer is often used to describe instruments that measure a continuous variable of a phenomenon where the spectral components are somehow mixed.

In visible light, spectrometer can separate white light and measures individual narrow bands of colour, called a spectrum. Furthermore, spectrometer such as mass spectrometer measures the spectrum of the masses of the atoms or molecules present in a gas. Spectrometers are used in Astronomy to analyse the chemical composition of stars and planets. It also gathers data on the origin of the universe.

Chromatic aberration: is also called chromatic distortion. It is the failure of a lens to focus all colours to the same point. It is caused by dispersion i.e. the refractive index of the lens elements varies with the wavelength of light. The refractive index of most transparent materials decreases with increasing wavelength.

Since, the focal length of a lens depends on the refractive index, this variation in refractive index affect focusing. Chromatic aberration manifests itself as 'fringes' of colour along boundaries that separate dark and bright parts of the image.

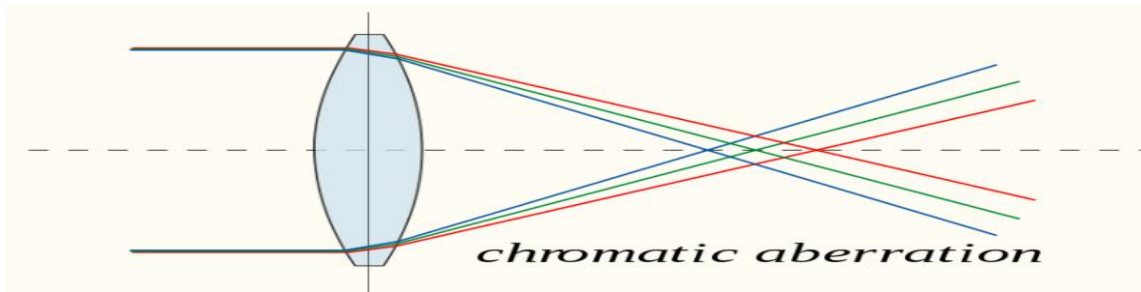


Fig. 17 The ray diagram of chromatic aberration.

2. The string produces a train of sinusoidal wave pulses on it, which is driven at a frequency of 3.00 Hz. the amplitude of the motion is 18.00 cm, and the wave speed is 30.00 m/s. (i.) Determine the angular frequency, ω and angular wave number, k for this wave. (ii.) Write an expression for the wave function. (iii.) Calculate the maximum values for the transverse speed and transverse acceleration of any point on the string.

Soln.

$$f = 3.00\text{Hz}, A = 18.0\text{cm} = 0.18\text{m}, V = 30.0\text{m/s}$$

i. Angular frequency, $\omega = \frac{2\pi}{T} = 2\pi f = 2\pi(3.00) = 18.85\text{rads}^{-1}$

Angular wave number, $k = \frac{\omega}{v} = \frac{18.85}{30.0} = 0.63\text{radm}^{-1}$

ii. The wave function; $y = A \sin(kx - \omega t)$

$$y = 0.18 \sin(0.63x - 18.85t)$$

iii. Maximum values for the transverse speed, $V_{y,max} = \omega A = 18.85 \times 0.18 = 3.393$

$$V_{y,max} = 3.39\text{m/s}$$

Maximum values for the transverse acceleration, $a_{y,max} = \omega^2 A = (18.85)^2 \times 0.18 = 63.95805$

$$a_{y,max} = 63.96\text{m/s}^2$$

3. A transverse wave on a taut string is modelled with the wave function:

$$y(x, t) = A \sin(kx - \omega t) = 0.2\text{m} \sin(6.28\text{m}^{-1}x - 1.57\text{s}^{-1}t).$$

Find the amplitude, wavelength, period and speed of the wave.

Soln:

i. Amplitude, $A = 0.2\text{m}$

ii. Wavelength, $\lambda = ?$

But Angular wave number, $k = \frac{2\pi}{\lambda}$.

$$\lambda = \frac{2\pi}{k}, \text{ where } k = 6.28\text{radm}^{-1}$$

$$\lambda = \frac{2\pi \text{rad}}{6.28 \text{radm}^{-1}} = 1.0 \text{m}$$

$$\lambda = 1.0 \text{m}$$

iii. Period, $T = ?$

But Angular frequency, $\omega = \frac{2\pi}{T}$.

$$T = \frac{2\pi}{\omega}, \text{ where } \omega = 1.57 \text{rads}^{-1}$$

$$T = \frac{2\pi \text{rad}}{1.57 \text{rads}^{-1}} = 4 \text{s}$$

$$T = 4 \text{s}.$$

iv. Speed of the wave, $V = \frac{\omega}{k} = \frac{1.57 \text{rads}^{-1}}{6.28 \text{radm}^{-1}} = 0.25$

$$V = 0.25 \text{m/s}.$$

EXERCISE:

1. Two waves are described by: $y_1 = 0.30 \sin[\pi(5x - 200t)]$

$$y_2 = 0.30 \sin\left[\pi(5x + 200t) + \frac{\pi}{2}\right]$$

Where y_1, y_2 and x are in meters and t is in seconds. When these two waves are combined, a traveling wave is produced. What are the (a) Amplitude, (b) Wave speed, and Wavelength of that traveling wave?

2. A wave on a string is described by, $y(x, t) = 15.0 \sin\left(\frac{\pi}{8}x - 4\pi t\right)$

Where x and y are in centimetres and t is in seconds.

(a) What is the transverse speed for a point on the string at $x = 6.00 \text{ cm}$ when $t = 0.250 \text{ s}$?

(b) What is the maximum transverse speed of any point on the string?

(c) What is the magnitude of the transverse acceleration for a point on the string at $x = 6.00 \text{ cm}$ when $t = 0.250 \text{ s}$?

(d) What is the magnitude of the maximum transverse acceleration for any point on the string?

1. An electric spark jumps along a straight line of length, $l = 10 \text{ m}$ emitting a pulse of sound that travels radially outward from the spark. (The spark is said to be a line source of sound). The power of this acoustic emission is $P_s = 1.5 \times 10^4 \text{ W}$.

a. What is the intensity, I of the sound when it reaches a distance $r = 120 \text{ cm}$ from the spark?

- b. At what time rate P_d is sound energy intercepted by an acoustic detector of area $A_d = 2.0 \text{ cm}^2$, aimed at the spark and located at distance, $r = 12 \text{ m}$ from the spark?
2. A point source emits sound waves isotropically. The intensity of the waves 2.50 m from the source is $1.91 \times 10^{-4} \text{ W/m}^2$. Assuming that the energy of the waves is conserved, find the power of the source.
 3. Propagation of sound in different media which are? Name them -----
 4. Find the speed of sound in an Aluminium bar whose modulus is $7.0 \times 10^{10} \text{ N/m}^2$ for Aluminium and its density is $2.70 \times 10^3 \text{ kg/m}^3$.
 5. What is the bulk modulus of a distilled water, if 32.0 g of distilled water occupies 22.4 litres and the speed of the sound in the distilled water is 1480 m/s .
 6. Two tuning forks of frequency 206 Hz and 194 Hz are sounded simultaneously. Calculate the frequency of the resultant beat.
 7. List Disadvantages of Echoes ----- and -----
 8. A stationary motion detector sends sound waves of frequency 0.150 MHz towards a truck approaching at a speed of 45.0 m/s . what is the frequency of the waves reflected back to the detector?